

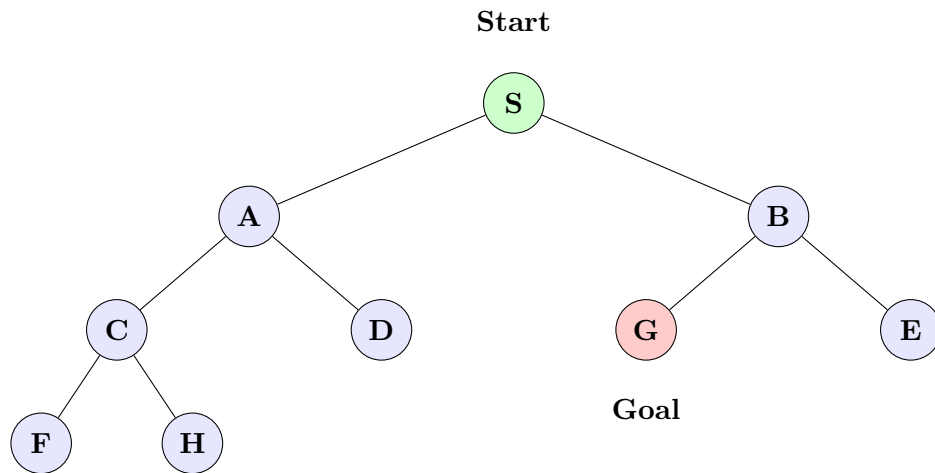
CS471 – Lesson 5 In-Class Worksheet

Uninformed Search (DFS, BFS, UCS)

Guided In-Class Worksheet (Instructor-Led)

Objective: Understand Search Space parameters, trace blind search algorithms (DFS/BFS), and execute Uniform Cost Search (UCS).

Scenario 1: A rescue drone is navigating a decision tree to find a lost squad (Goal ‘G’).

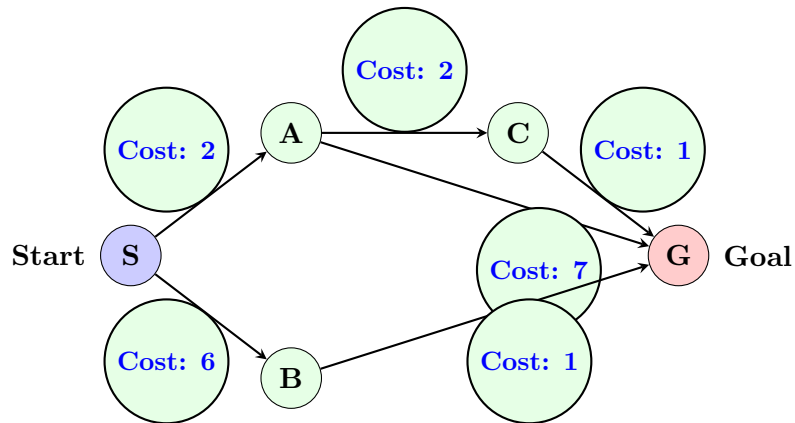


Part 1: Search Space Parameters & Uninformed Trees

1. **Branching Factor (b):** What is the maximum number of successors from any node in this tree?
2. **Maximum Depth (m):** What is the absolute maximum depth of this search tree?
3. **Depth of Shallowest Solution (s):** At what depth level is the easiest, shortest path to the goal?
4. **Depth-First Search (DFS):** DFS uses a LIFO Stack. List the exact sequence of nodes the drone will *expand* (visit and evaluate) before successfully finding the Goal.

5. **Breadth-First Search (BFS):** BFS uses a FIFO Queue. List the exact sequence of nodes the drone will *expand* before successfully finding the Goal.

Scenario 2: A convoy must route through a map where roads vary in terrain. The convoy must find the absolutely optimal (cheapest) route to the Goal.



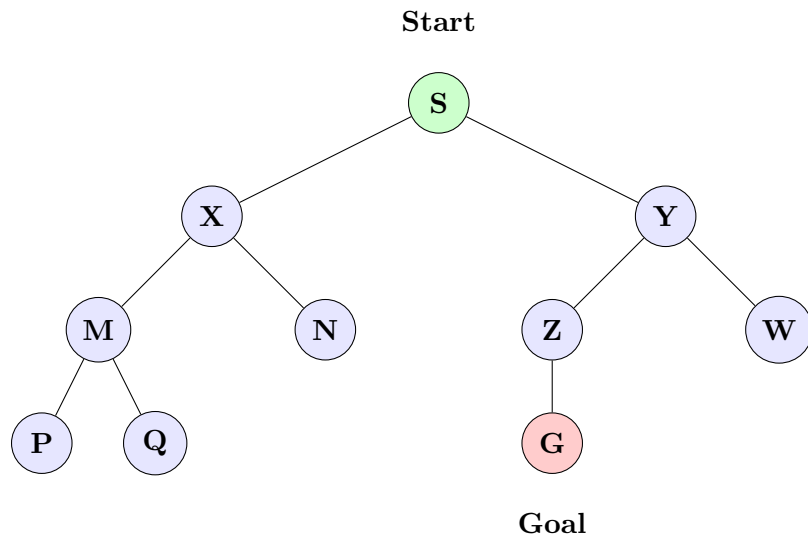
Part 2: Uniform Cost Search (UCS)

6. **Trace the Priority Queue:** Trace the execution of UCS. At each step, list the contents of the frontier (the priority queue) and circle which node is expanded next. Track the cumulative path cost $g(n)$ from the Start node.
7. **Final Path & Optimality:** What is the final path selected by UCS, and what is its total cost?

Student Practice

Objective: Independently trace parameters, DFS, BFS, and UCS on novel environments.

Scenario 3: A surveillance drone must navigate the following communication tree to locate a specific signal broadcast (Goal 'G').

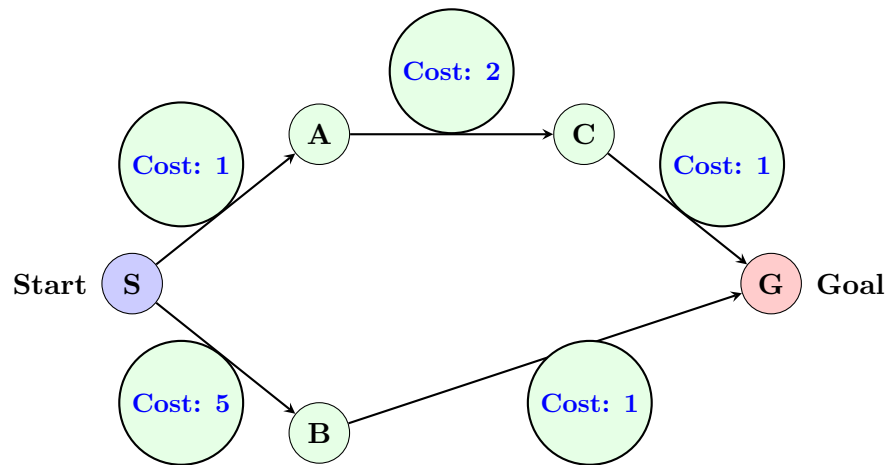


Part 3: Uninformed Trees Practice

8. **Depth-First Search (DFS) Trace:** List the exact sequence of nodes the drone will *expand* before successfully finding the Goal.

9. **Breadth-First Search (BFS) Trace:** List the exact sequence of nodes the drone will *expand* before successfully finding the Goal.

Scenario 4: A ground extraction team must navigate a hostile urban grid. Moving down main avenues is fast (Cost 1), but moving through alleyways is dangerous and slow (Cost 5).



Part 4: Uniform Cost Search (UCS) Practice

10. **Trace the Queue:** Trace the execution of Uniform Cost Search (UCS). At each step, list the contents of the frontier and circle the node expanded next.
11. **Final Path & Breadth-First Flaw:** What is the final path selected by UCS, and what is its total cost? Why would standard Breadth-First Search (BFS) fail to find this optimal path?